

PROJECT DELIVERABLE - PHASE 1

Ideation Phase

Problem Statement, Empathy Mapping, Ideation & Feasibility Analysis

Platform: ServiceNow PDI dev423661 | Global Scope | September 2026

1. Project Title & Executive Summary

Project Title: Metro Ticket Generating System using ServiceNow (dev423661.service-now.com)

The Metro Ticket Generating System is an enterprise cloud application engineered on the ServiceNow Personal Developer Instance (PDI) to digitize urban transit ticketing. By replacing manual ticketing counters with a responsive self-service portal, the system provides contactless ticket booking, automated dynamic fare calculation, and instant scannable QR ticket issuance, delivering a zero-queue commuter experience.

2. Problem Statement & Background

Urban rapid transit networks experience extreme passenger congestion at physical ticket token vending booths during peak commute hours. The traditional counter-based ticketing model presents significant operational bottlenecks and commuter friction:

- Long Commuter Queues: Physical ticketing windows create 15 to 30 minute delays during morning and evening rush hours.
- Manual Calculation Errors: Station clerks manually consult static fare charts, leading to calculation discrepancies and disputes.
- Cash & Paper Dependency: Paper-based slips and plastic tokens result in physical material waste and recurring operational replenishment costs.
- Absence of Self-Service: Lack of an online self-service interface forces all commuters through station gate choke points.

3. Empathy Map Canvas

An empathy mapping study was conducted across daily metropolitan transit commuters to capture primary behavioral dimensions:

Perspective	Commuter Observation & Feedback
Says	'The queue moves too slowly; I need a fast way to book my return trip..
Thinks	'Why cannot I book directly from my phone or web browser and scan my ..
Does	Rushes to counter, searches for exact currency change, stores paper r..
Feels	Frustrated by peak-hour transit bottlenecks, anxious about missing co..
Pains	Time lost in ticket lines, currency exchange hurdles, loss of printed..
Gains	Zero-wait ticketing, instant dynamic fare computation, contactless QR..

4. Ideation & Proposed Solution

To resolve the identified commuter pains, the team ideated a cloud-native ServiceNow digital ticketing application leveraging Service Catalog, client-side GlideForm scripts, conditional UI policies, custom data tables, and an AngularJS QR modal widget:

- Service Portal Catalog Item: Commuters navigate to the self-service catalog (/sp) to access the 'Book A Metro Ticket' form.
- Dynamic Station Lookups: Reference variables connect to a master data table storing verified metro stations (Ameerpet, Madhapur, etc.).
- Automated Fare Scripting: Client-side onChange scripts compute journey fares dynamically without requiring manual server round-trips.

- Conditional Payment Visibility: UI Policy rules show 'Enter payment mode' exclusively when 'Others' is selected as the payment method.
- Instant QR Modal Delivery: Custom Service Portal Widget displays a scannable digital QR code instantly before submitting the order.

5. Existing vs. Proposed System Comparison

Feature Dimension	Existing Legacy Ticketing	Proposed ServiceNow System
Ticketing Channel	Physical station counter / token ..	Digital ServiceNow Service Portal (..
Fare Computation	Manual lookup from printed charts	Automated JavaScript client-side sc..
Ticket Medium	Printed thermal paper slip or token	Digital QR code rendered in modal d..
Payment Support	Predominantly cash or static card..	UPI, Debit/Credit Card, and Custom ..
Record Persistence	Fragmented local station database	Centralized sc_req_item / RITM tabl..
Environmental Imp..	High paper waste and discarded re..	100% paperless, zero physical waste

6. Feasibility Study

Technical Feasibility

The project is fully viable using ServiceNow Personal Developer Instance (Australia release) with Service Catalog, Glide Form APIs, spModal framework, and public REST QR Server API. All required dependencies execute natively in modern web browsers without native mobile wrappers.

Economic & Operational Feasibility

ServiceNow developer instances are cost-free for prototyping. In an enterprise deployment, digitizing 70% of tickets saves substantial paper, hardware maintenance, and staffing costs. Commuters access the portal immediately without specialized terminal hardware.

7. Phase 1 Deliverables & Sign-Off

Deliverables Completed: Problem Statement Document, Commuter Persona & Empathy Map, Solution Ideation Architecture, Feasibility Report.